

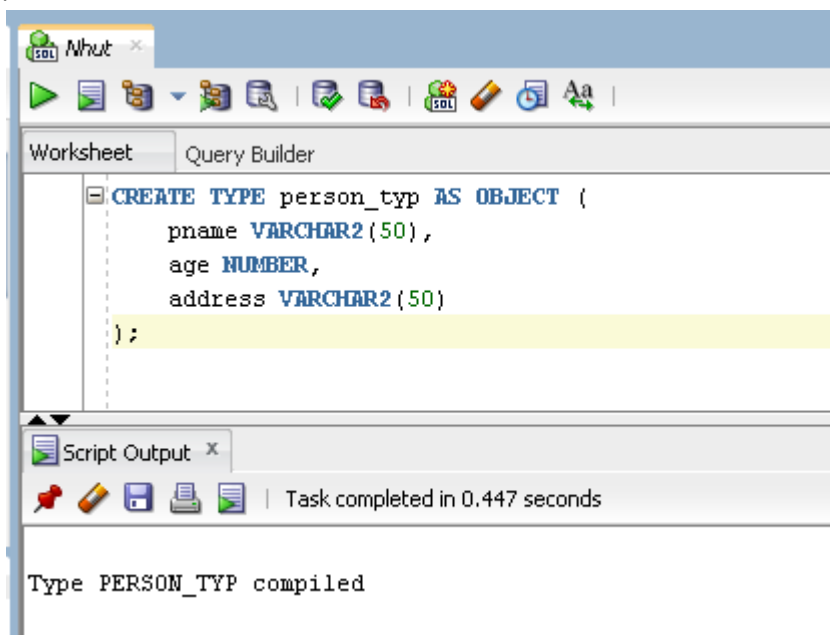
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Lab6. Oracle object-relational database

1. Write a PL/SQL code to create a class for a "Person" with attributes such as name, age, and address.

Code:

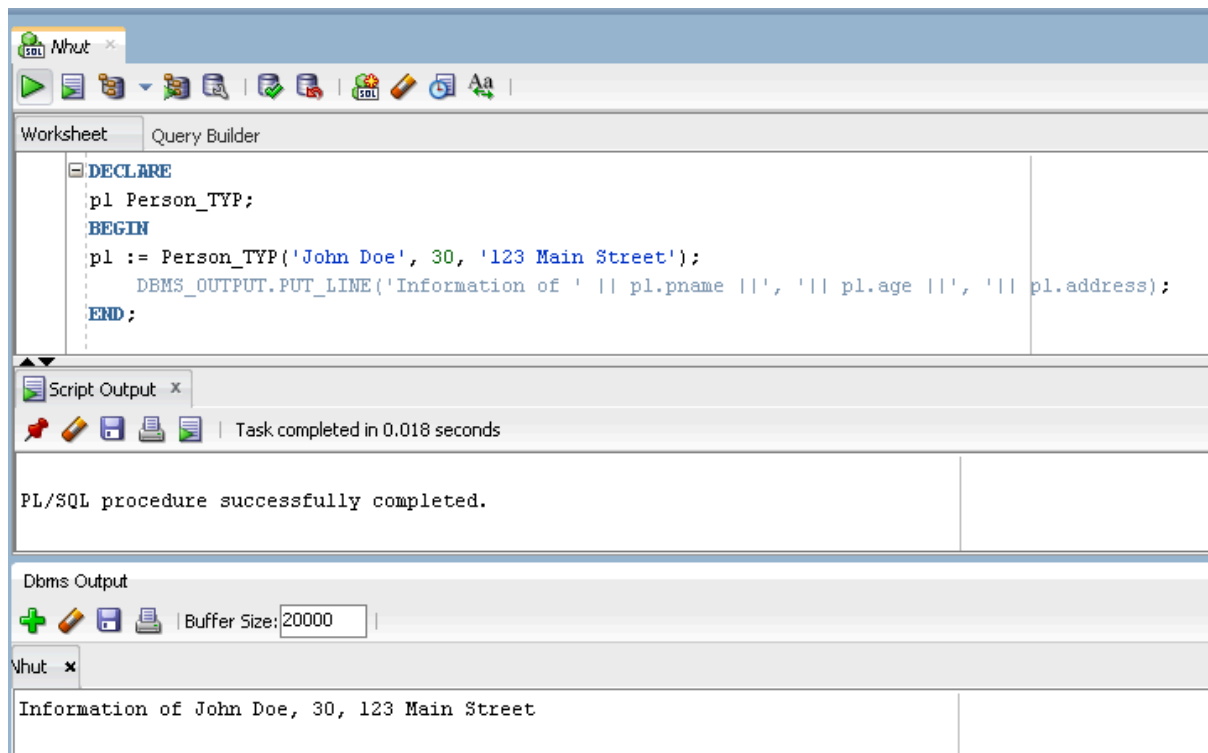
```
CREATE TYPE person_typ AS OBJECT (  
    pname VARCHAR2(50),  
    age NUMBER,  
    address VARCHAR2(50)  
);
```



2. Create an object p1 of type Person with the following attribute values: ('John Doe', 30, '123 Main Street') and display its attributes to the screen.

Code:

```
DECLARE  
p1 Person_TYP;  
BEGIN  
p1 := Person_TYP('John Doe', 30, '123 Main Street');  
    DBMS_OUTPUT.PUT_LINE('Information of ' || p1.pname || ', ' || p1.age || ', ' ||  
p1.address);  
END;
```



3. Write a PL/SQL code to implement methods in the "Person" class to display the details and update the age.

Code:

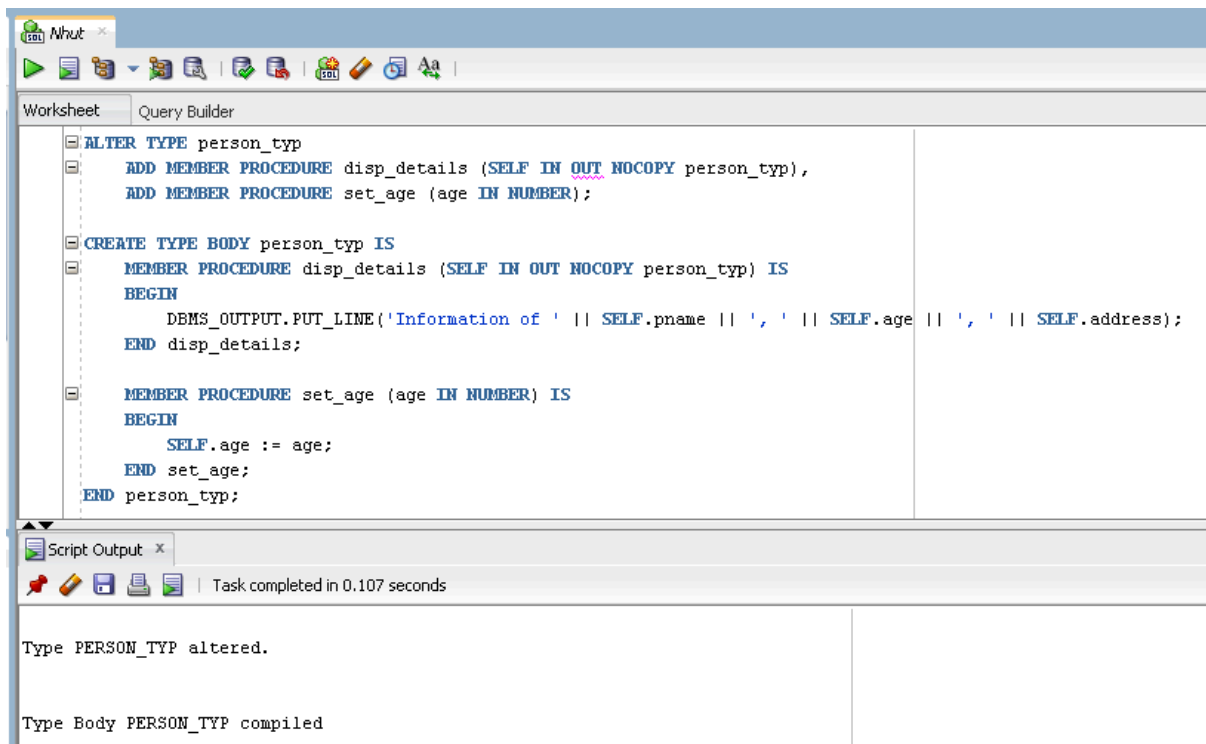
```
ALTER TYPE person_typ
```

```
ADD MEMBER PROCEDURE disp_details (SELF IN OUT NOCOPY person_typ),
ADD MEMBER PROCEDURE set_age (age IN NUMBER);
```

```
CREATE TYPE BODY person_typ IS
```

```
MEMBER PROCEDURE disp_details (SELF IN OUT NOCOPY person_typ) IS
BEGIN
DBMS_OUTPUT.PUT_LINE('Information of ' || SELF.pname || ', ' || SELF.age || ', ' || SELF.address);
END disp_details;
```

```
MEMBER PROCEDURE set_age (age IN NUMBER) IS
BEGIN
SELF.age := age;
END set_age;
END person_typ;
```



4. Write a PL/SQL code to implement a method to calculate the annual bonus based on the

salary in the "Employee" class, given:

salary < 50000: bonus_percentage := 0.1;

salary < 100000: bonus_percentage := 0.15;

otherwise: bonus_percentage := 0.2;

bonus_amount = salary * bonus_percentage;

Code:

```

CREATE TYPE Employee AS OBJECT (
    ename VARCHAR2(50),
    salary NUMBER,
    MEMBER FUNCTION calculate_bonus RETURN NUMBER
);

```

```

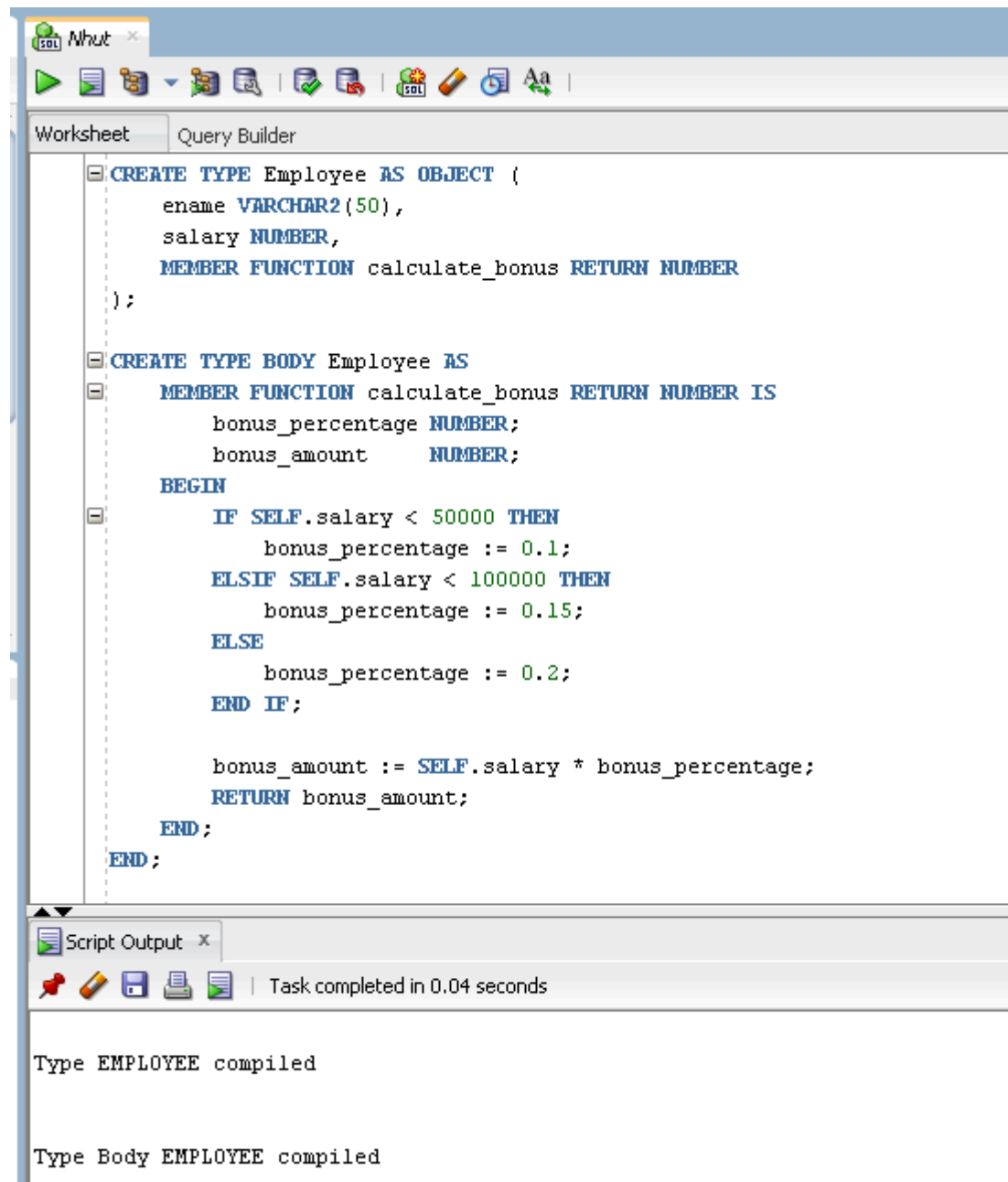
CREATE TYPE BODY Employee AS
MEMBER FUNCTION calculate_bonus RETURN NUMBER IS
    bonus_percentage NUMBER;
    bonus_amount    NUMBER;
BEGIN
    IF SELF.salary < 50000 THEN
        bonus_percentage := 0.1;
    ELSIF SELF.salary < 100000 THEN
        bonus_percentage := 0.15;
    ELSE
        bonus_percentage := 0.2;
    END IF;

```

```

        bonus_amount := SELF.salary * bonus_percentage;
    RETURN bonus_amount;
END;
END;

```



5. Write a PL/SQL code to create a "Shape" class with attributes like length and width and implement a method to calculate the area.

Code:

```

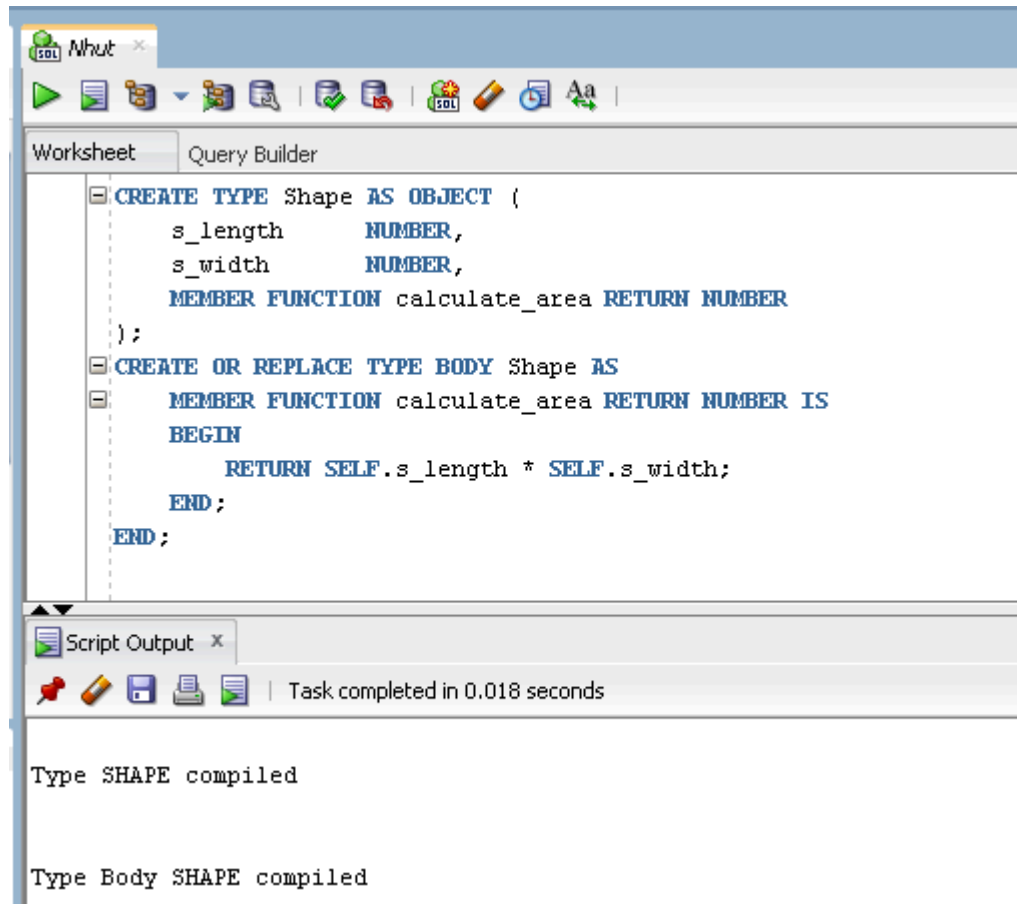
CREATE TYPE Shape AS OBJECT (
    s_length    NUMBER,
    s_width     NUMBER,
    MEMBER FUNCTION calculate_area RETURN NUMBER
);

```

```

CREATE OR REPLACE TYPE BODY Shape AS
    MEMBER FUNCTION calculate_area RETURN NUMBER IS
    BEGIN
        RETURN SELF.s_length * SELF.s_width;
    END;
END;

```

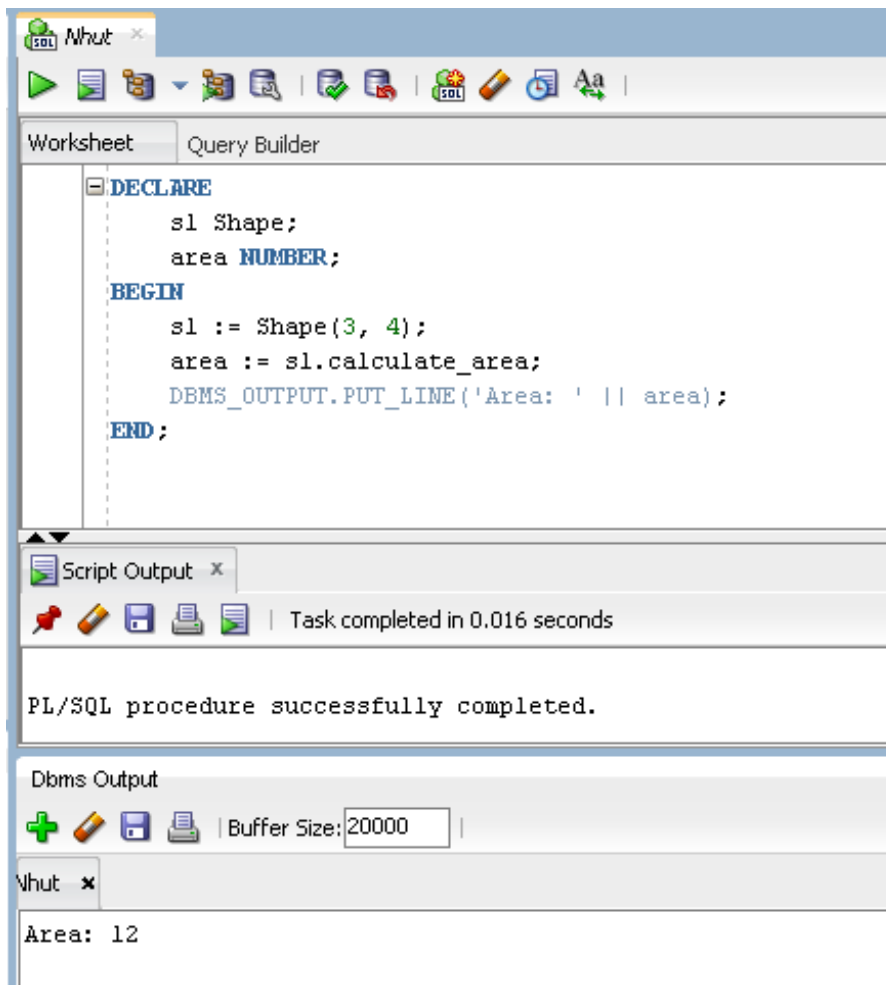


Test:

```

DECLARE
    s1 Shape;
    area NUMBER;
BEGIN
    s1 := Shape(3, 4);
    area := s1.calculate_area;
    DBMS_OUTPUT.PUT_LINE('Area: ' || area);
END;

```



6. Write a PL/SQL code that implements a method in the "Shape" class to determine whether the shape is square

Code:

ALTER TYPE Shape

ADD MEMBER FUNCTION is_square RETURN BOOLEAN;

CREATE OR REPLACE TYPE BODY Shape AS

MEMBER FUNCTION is_square RETURN BOOLEAN IS

BEGIN

IF SELF.length = SELF.width THEN

RETURN TRUE;

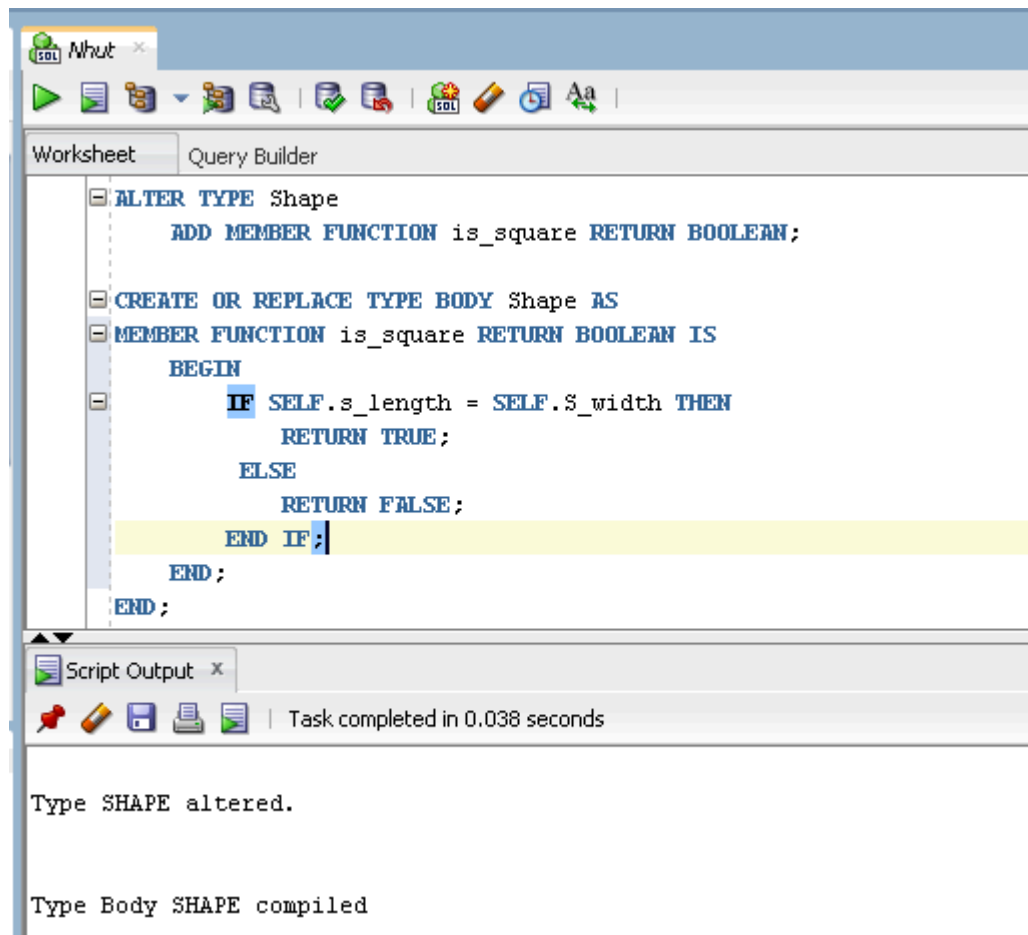
ELSE

RETURN FALSE;

END IF;

END;

END;



Test:

Nhut x

Worksheet Query Builder

```
DECLARE
s Shape;
result BOOLEAN;
BEGIN
s := Shape(10, 10);
result := s.is_square;
IF result THEN
DBMS_OUTPUT.PUT_LINE('Square ?: TRUE');
ELSE
DBMS_OUTPUT.PUT_LINE('Square ?: FALSE');
END IF;
END;
```

Script Output x

Task completed in 0.019 seconds

PL/SQL procedure successfully completed.

Dbms Output

Buffer Size: 20000

Nhut x

Square ?: TRUE

Nhut x

Worksheet Query Builder

```
DECLARE
s Shape;
result BOOLEAN;
BEGIN
s := Shape(10, 5);
result := s.is_square;
IF result THEN
DBMS_OUTPUT.PUT_LINE('Square ?: TRUE');
ELSE
DBMS_OUTPUT.PUT_LINE('Square ?: FALSE');
END IF;
END;
```

Script Output x

Task completed in 0.017 seconds

PL/SQL procedure successfully completed.

Dbms Output

Buffer Size: 20000

Nhut x

Square ?: FALSE